

PKCERT AI & Software Development Internship

Task 14 – Ensemble Methods: Bagging & Boosting

Objective:

Develop practical skills in implementing ensemble learning techniques, including Bagging and Boosting. Gain introductory experience with XGBoost and LightGBM.

Total Marks: 100

Part A – Dataset Selection & Preparation (15 Marks)

1. Select a public classification dataset.
2. Describe the dataset, identify features and target variable.
3. Preprocess the dataset and split it into training and testing sets.

Part B – Bagging (30 Marks)

1. Implement a Bagging classifier.
2. Evaluate using Accuracy, Precision, Recall, F1-Score, and Confusion Matrix.
3. Explain the advantages, limitations, and use cases of Bagging.

Part C – Boosting (40 Marks)

1. Implement an XGBoost or LightGBM classifier.
2. Evaluate the model using standard classification metrics.
3. Compare the Boosting model with the Bagging model.
4. Explain the advantages, limitations, and applications of Boosting.

Part D – Comparative Analysis & Recommendation (15 Marks)

1. Compare Bagging and Boosting based on performance, computational efficiency, and suitability for the selected dataset.
2. Recommend the most appropriate ensemble method with justification.

Assessment Area	Marks
Dataset Preparation	15
Bagging	30
Boosting (XGBoost/LightGBM)	40

Comparative Analysis & Recommendation	15
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Total	100
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